

EchoSafe -- ICS-43434 Mic Footprint Fit-Test Template

Confirm this printout is at TRUE 1:1 SCALE before trusting any alignment below.

Most printers/PDF viewers silently rescale ('fit to page') -- measure the 100mm ruler with a real ruler first.



0mm

100mm -- this whole line must measure exactly 100mm with a real ruler.

How to use:

1. Print at 100% / Actual Size -- NOT 'Fit to page' or 'Shrink to fit'.
2. Measure the ruler above with a real ruler. If it isn't exactly 100mm, redo the print with scaling disabled.
3. The drawing below (labelled 1:1 SCALE) is MIC1's real pad layout + GND ring + the H1 acoustic port hole, at the exact position they sit at on the real board (verified against EchoSafe_FrontLeft.kicad_pcb: both footprints share the same origin coordinate, no rotation).
4. At this part's scale (~3.5 x 2.65mm), print/scan tolerance alone can exceed what visual alignment can verify -- use CALIPERS against the real ICS-43434 part for the actual check, not just eyeballing the print.
5. Pin-name callouts (WS/LR/GND/SCK/VDD/SD) are for identification only. This PDF is vector, so on-screen you can zoom in arbitrarily far in any PDF viewer without losing sharpness -- no separate enlarged copy needed.
6. If pad positions don't match: measure the real offset and edit
hardware/EchoSafe_RevA/lib/symbols/Mics_Corrected.pretty/ICS-43434_LGA6.kicad_mod
(same pattern as every other footprint fix in this project -- see CLAUDE.md > Hardware > Footprints).

MIC1 + H1 -- 1:1 SCALE (use calipers, not the eye)

v real footprint instance, positioned below

MIC1 (1:1 SCALE)

